

Full-Custom Design of a CMOS Operational Transconductance Amplifier (OTA)

Analog/RF IC Design, Full-Custom Layout and Post-Layout Verification Portfolio

Topics: CMOS OTA · Current-Mirror OTA · Differential Pair · Transistor Biasing · Small-Signal AC Analysis · DC Operating Point · Slew Rate · Settling Time · Common-Mode Range · Output Swing · Physical Layout · DRC/LVS · Post-Layout Verification · Silicon-Validation Methodology

Tools: Cadence Virtuoso

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Abstract

This report presents the complete full-custom design of a CMOS Operational Transconductance Amplifier (OTA) in a 180 nm-class 2 V CMOS process using Cadence Virtuoso. A single-stage symmetrical current-mirror topology is adopted, comprising a differential input pair, PMOS current-mirror loads, NMOS output mirrors, a tail current source, and a separate bias-current generator. The design is taken through the entire analog integrated-circuit flow: transistor-level schematic capture; small-signal AC analysis for gain, unity-gain bandwidth, and phase margin; DC operating-point verification; large-signal transient (slew-rate and settling) characterization; input/output common-mode range analysis; matching-aware physical layout; design-rule check (DRC) and layout-versus-schematic (LVS) sign-off; full-circuit integration; and post-layout simulation with extracted parasitics. Each result is interpreted against closed-form analog design relations. The amplifier achieves a DC gain of ≈ 24.85 dB, a unity-gain bandwidth of ≈ 14.6 MHz, a phase margin of $\approx 73^\circ$, and a slew rate of ≈ 15.7 V/ μ s; both the OTA core and the bias network are DRC- and LVS-clean, and post-layout simulation confirms the schematic-level performance.

Index Terms- Operational transconductance amplifier (OTA), CMOS analog design, current mirror, transconductance, dominant pole, phase margin, slew rate, common-mode range, Cadence Virtuoso, DRC, LVS, post-layout extraction, device matching.

Parameter	Value	Basis
DC gain, A_0	24.85 dB	Low-frequency marker, AC magnitude
Unity-gain bandwidth, f_u	14.6 MHz	0 dB magnitude crossover
Phase at f_u	-106.7°	Phase marker at crossover
Phase margin, PM	73°	$180^\circ + \angle A(f_u)$
Slew rate, SR	15.7 V/ μ s	Transient rising-edge dV/dt
DRC / LVS	0 / 0	Clean on both blocks
Post-layout	Meets spec	Parasitic-extracted netlist

Table 1. Key performance parameters of the OTA.

1. Introduction and Design Flow

An Operational Transconductance Amplifier (OTA) is a voltage-controlled current source whose output current is proportional to the differential input voltage through the small-signal transconductance g_m :

$$i_{out} = g_m (v_+ - v_-), \quad g_m = \sqrt{2 \mu_n C_{ox} (W/L) I_D} \quad (1)$$

where $\mu_n C_{ox}$ is the process transconductance parameter, W/L the device aspect ratio, and I_D the drain current. OTAs are the fundamental gain element of analog and mixed-signal CMOS systems- switched-capacitor filters, data converters, sensor front-ends, and continuous-time filters all depend on them. This work implements a single-stage symmetrical OTA and carries it through the full custom-IC design flow, with the goal of demonstrating end-to-end ownership of an analog block: not merely a functioning schematic, but a manufacturable, verification-clean physical design whose post-layout behaviour matches specification.

1.1 Design goals

The main design goal was to realize a stable single-stage CMOS OTA with sufficient DC gain and a unity-gain bandwidth in the tens of MHz range. The design was also intended to ensure reliable closed-loop operation by achieving a phase margin comfortably above the 60° stability target. In addition to schematic-level performance, the project focused on creating a symmetric and matching-aware physical layout using dummy devices, followed by verification against both the schematic through LVS and the foundry design rules through DRC. Finally, post-layout validation was performed to confirm that extracted layout parasitics did not degrade the amplifier performance below the intended specification.

1.2 Design flow

Table 2 lists each design activity and its purpose. The flow proceeds from schematic design, through small- and large-signal simulation, to physical implementation and parasitic-aware sign-off.

Design activity	Purpose
OTA schematic	Create the transistor-level gain cell
Small-signal AC analysis	Extract gain, bandwidth, and phase margin
DC operating point	Verify bias currents, gm, node voltages, and headroom
Transient analysis	Measure slew rate and settling response
Common-mode analysis	Define usable input and output voltage ranges
Layout + DRC/LVS	Verify manufacturability and schematic–layout equivalence
Post-layout simulation	Quantify the effect of extracted parasitics

Table 2. Design activities and their purpose in the analog IC flow.

2. Circuit Architecture and Design Equations

2.1 OTA core

The OTA core (Figure 1) is built around a differential input pair (M1, M6; 2 V NMOS, $W/L \approx 2.885 \mu\text{m} / 400 \text{ nm}$) biased by a tail current source (M7). The drain currents of the input pair are processed by PMOS current-mirror loads (M2–M5) across the top rail and steered into the output branch by NMOS mirrors (M8, M9), producing a single-ended output V_{out} . Each half of the differential signal is mirrored and recombined at the output node, so the effective transconductance is the input-pair g_{m1} scaled by the mirror ratio B .

To first order, the DC gain is the product of the effective transconductance and the output resistance at V_{out} , and the unity-gain bandwidth is set by that transconductance and the load capacitance:

$$A_0 = B g_{m1} (r_{o,p} \parallel r_{o,n}), \quad f_u = B g_{m1} / (2\pi C_L) \quad (2)$$

where $r_o = 1/(\lambda I_D)$ is the device output resistance. Because the topology has a single high-impedance node, it is dominant-pole compensated by the load itself; the dominant pole is

$$f_{p1} = 1 / (2\pi R_{\text{out}} C_L), \quad R_{\text{out}} = r_{o,p} \parallel r_{o,n} \quad (3)$$

Devices M10 and M11 are dummy transistors: they carry no signal but replicate the local layout environment of the active devices so that etch, mechanical-stress, and lithographic effects are matched across the critical pairs. Matching is decisive because the input-referred offset is dominated by threshold mismatch, which follows Pelgrom's law,

$$\sigma_{\Delta V_{\text{th}}} = A_{V_{\text{th}}} / \sqrt{WL} \quad (4)$$

so larger, well-matched devices reduce offset and even-order distortion.

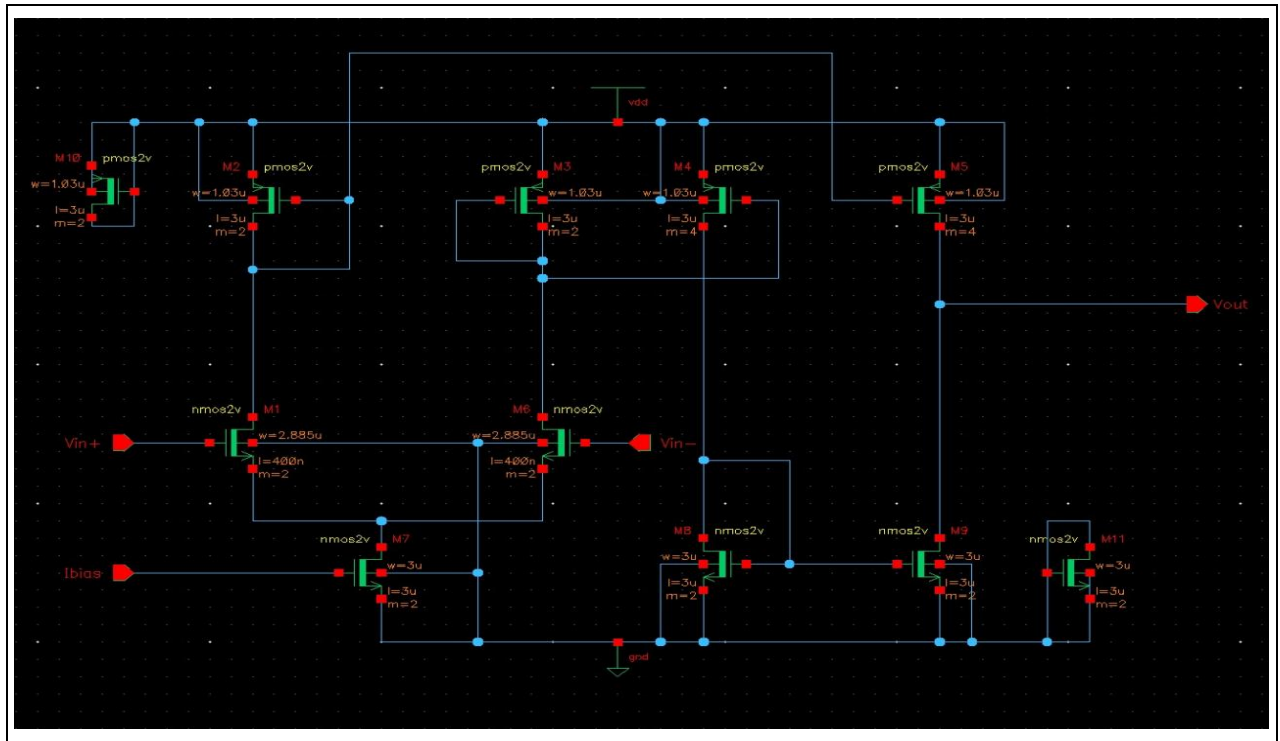


Figure 1. OTA core schematic- differential pair (M1/M6), PMOS mirror loads (M2–M5), output mirrors (M8/M9), tail source (M7); dummy transistors M10, M11.

The schematic shows the complete single-stage gain cell. The differential pair M1/M6 converts the differential input into a drain-current imbalance $\Delta i = g_{m1}V_{id}$, which the PMOS mirrors M2–M5 fold and the NMOS mirrors M8/M9 recombine into a single-ended current at V_{out} . Because all gain is developed at one high-impedance node, the stage is inherently dominant-pole compensated and needs no Miller capacitor, giving the gain and bandwidth of Eq. (2). The price of this simplicity is a moderate, non-cascoded gain of order $g_m r_o$; if higher gain were required, the loads M2–M5 would be cascoded at the cost of output swing. The deliberately large input-device length (400 nm rather than minimum) raises r_o and lowers $1/f$ noise and mismatch, a typical analog trade of speed for accuracy.

2.2 Figures of merit and analytical relations

Beyond DC gain and bandwidth, four further relations define the quality of the amplifier and are used to interpret the simulated results. First, because the stage has a single dominant pole, the gain–bandwidth product is constant and equal to the unity-gain frequency:

$$GBW = A_0 \cdot f_{p1} = B g_{m1} / (2\pi C_L) = f_u \quad (5)$$

Second, the common-mode rejection ratio (CMRR) measures how strongly the amplifier discriminates the wanted differential signal against a common-mode disturbance. It is set by the output resistance of the tail current source $r_{o,tail}$, which converts a common-mode input into a (rejected) tail-current variation:

$$CMRR = A_{dm} / A_{cm} \approx 2 g_{m1} r_{o,tail} \cdot (r_{o,p} \parallel r_{o,n}) \quad (6)$$

a higher tail output resistance (a longer or cascoded tail device) directly improves CMRR. Third, the power-supply rejection ratio (PSRR) describes how supply noise couples to the output; for the positive rail it is approximately the ratio of the differential gain to the supply-to-output gain,

$$\text{PSRR}^+ = A_{\text{dm}} / A_{\text{vdd}}, \quad A_{\text{vdd}} \approx g_{\text{m,p}} (r_{\text{o,p}} \parallel r_{\text{o,n}}) \quad (7)$$

Fourth, the input-referred thermal noise of the OTA is dominated by the input pair and the active-load devices; referring the channel noise of all four contributors to the input gives the input-referred noise spectral density

$$\overline{v_{\text{n,in}}^2} = (8kT / 3g_{\text{m1}}) [1 + g_{\text{m,p}}/g_{\text{m1}}] \quad (\text{V}^2/\text{Hz}) \quad (8)$$

which shows that maximizing the input g_{m1} relative to the load $g_{\text{m,p}}$ minimizes noise — the same direction that benefits offset and CMRR, and the reason the input pair is biased at the highest practical g_{m} . Equations (5)–(8) are the analytical background; the verified numerical results in this report are the gain, bandwidth, phase margin, slew rate, operating point, common-mode range, DRC/LVS, and extracted post-layout behavior shown in the Cadence results.

2.3 Bias network

A dedicated bias circuit (Figure 2) generates the reference current I_{bias} that sets the tail current and hence the operating point of every branch. It is a PMOS/NMOS mirror chain distributing a stable, supply-tracking current; dummy devices M10–M12 preserve symmetry in the mirror legs. Because $g_{\text{m}} \propto \sqrt{I_{\text{D}}}$ (Eq. 1), the gain, bandwidth, slew rate, and swing all track I_{bias} , so a stable reference is essential. Designing the bias network as a separate, independently verified block keeps the OTA core reusable.

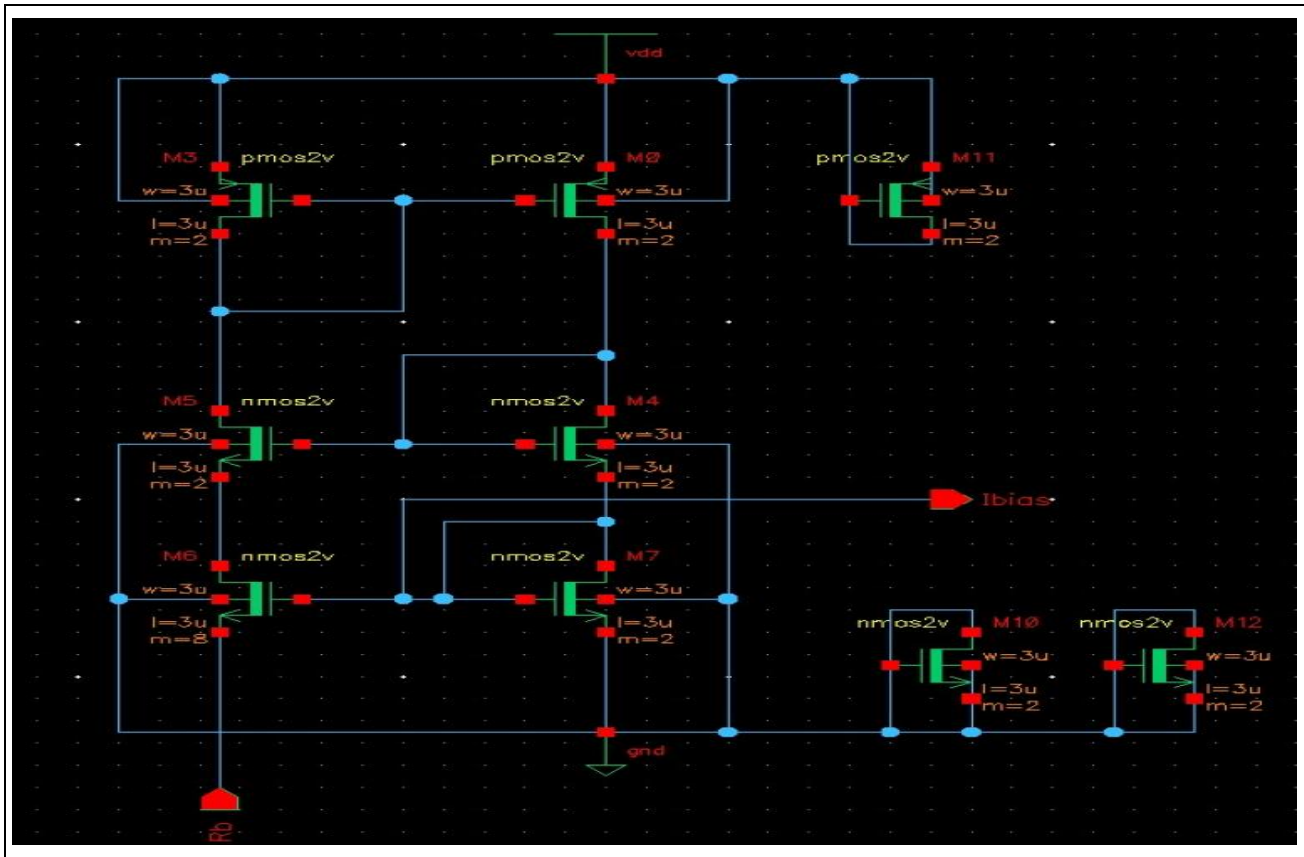


Figure 2. Bias-current generation circuit with dummy devices M10, M11, M12.

The schematic shows the standalone bias generator. It establishes I_{bias} through a matched PMOS/NMOS mirror chain and replicates it into the OTA tail, so the entire amplifier inherits one well-defined reference. Mirror accuracy is limited by device matching and by channel-length modulation, which makes the copied current depend weakly on the output voltage; lengthening the mirror devices raises their r_o and suppresses this dependence. Because every key OTA metric scales with I_{bias} — $g_m \propto \sqrt{I_{bias}}$, bandwidth $\propto g_m$, and slew rate $= I_{tail}/C_L$ — the reference current is the single most influential design variable, which is why it is generated and verified as an independent block before integration.

2.4 Suitability of the topology

The single-stage symmetrical OTA trades the very high DC gain of a cascode or two-stage Miller design for simplicity, inherent single-pole stability, and a compact, easily-matched layout. With no Miller compensation capacitor to size, stability is governed by the output-node pole of Eq. (3) — consistent with the $\approx 73^\circ$ phase margin measured in Section 3.

3. Small-Signal (AC) Verification

Open-loop AC analysis characterizes the two quantities that define usefulness in feedback: gain and stability. The test bench (Figure 3) first establishes the DC operating point, then applies a small-signal differential stimulus and sweeps frequency. With a single dominant pole, the loop phase at the unity-gain frequency determines the phase margin:

$$PM = 180^\circ + \angle A(f_u), \quad A(f) = A_0 / (1 + jf/f_{p1}) \quad (9)$$

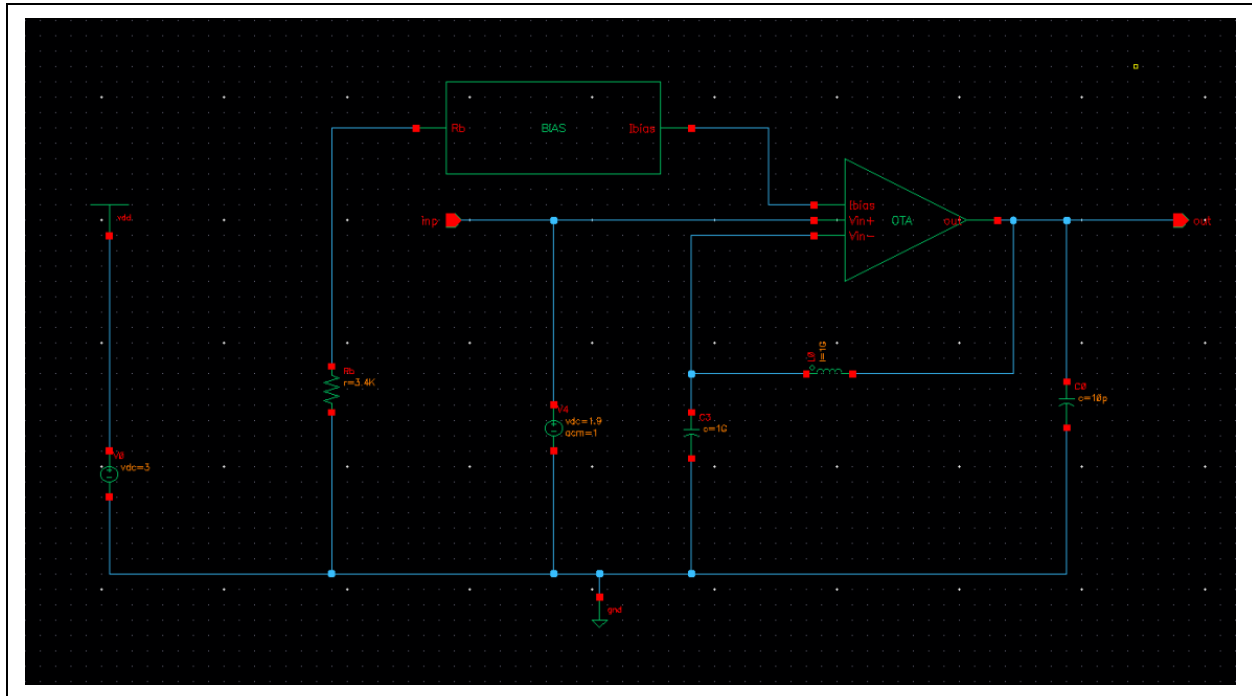


Figure 3. AC analysis test bench-OTA in open-loop configuration with the bias block, supply, load capacitor, and small-signal stimulus.

The diagram shows the open-loop AC test bench. The OTA symbol is connected to its bias block, supply, and load capacitor, and a small-signal differential source drives the inputs while the DC operating point is held fixed. Sweeping frequency on this configuration yields the open-loop magnitude and phase directly, from which gain, unity-gain frequency, and phase margin (Eq. 9) are read. Testing open-loop- rather than in a closed loop- is deliberate: it exposes the raw $A(f)$ of the amplifier, so the stability margin can be judged before any feedback network is committed.

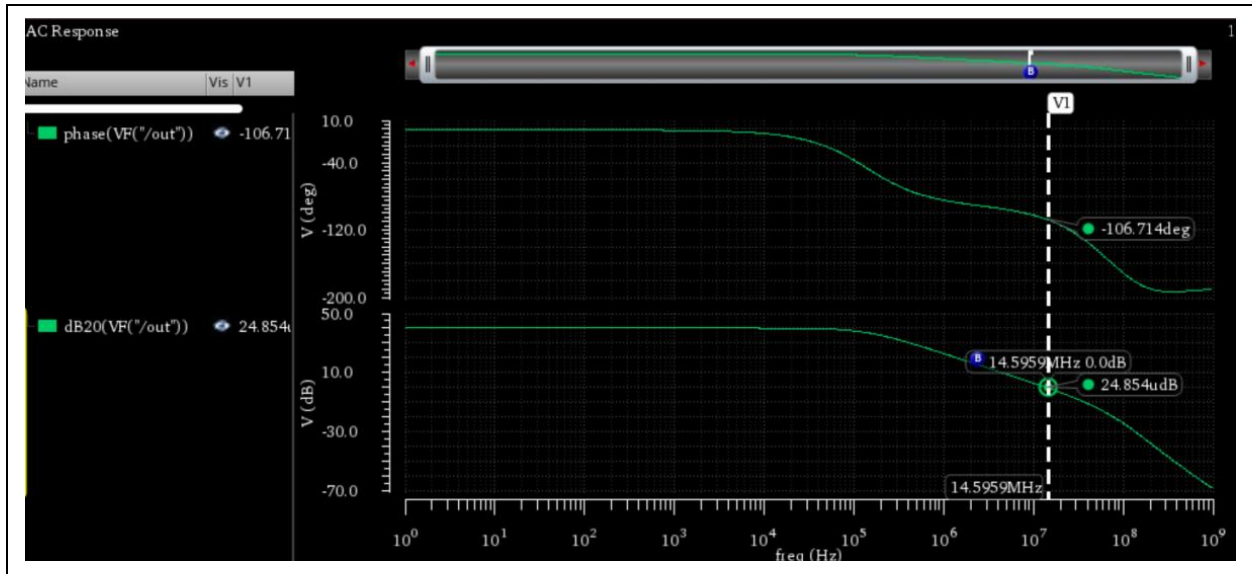


Figure 4. AC response- magnitude (dB20) and phase. Magnitude crosses 0 dB at ≈ 14.6 MHz with phase $\approx -106.7^\circ$, giving phase margin $\approx 73^\circ$.

3.1 Results and analysis

AC metric	Observed value	Notes
Low-frequency gain, A_0	≈ 24.85 dB	Open-loop DC gain, Eq. (2)
Unity-gain bandwidth, f_u	≈ 14.6 MHz	Magnitude crosses 0 dB
Phase at f_u	$\approx -106.7^\circ$	From phase plot at crossover
Phase margin, PM	$\approx 73^\circ$	From Eq. (9)

Table 3. Small-signal AC results.

The plot shows the Bode magnitude (dB20) and phase. At low frequency the gain is flat at ≈ 24.85 dB, confirming the $g_m R_{out}$ product of Eq. (2); the magnitude then rolls off at -20 dB/decade- the signature of a single dominant pole — and crosses 0 dB at ≈ 14.6 MHz. The phase at that crossover is $\approx -106.7^\circ$, giving a phase margin of $\approx 73^\circ$ by Eq. (9). The single-slope roll-off and the absence of any magnitude peaking confirm that the output node alone sets the bandwidth and that the next (non-dominant) pole lies well beyond f_u ; a second pole near f_u would have eroded the phase margin and produced peaking. A 73° margin sits above the 60° textbook target, trading a little speed for a guaranteed overshoot-free closed-loop response.

3.2 Analytical verification against the operating point

To confirm that the simulated response follows from the device physics rather than a tool artifact, the small-signal results are reproduced by hand from the DC operating-point data of Figure 6. The annotated values are: input-pair transconductance $g_{m1} \approx 535 \mu\text{S}$ at $I_{D1} \approx 39.9 \mu\text{A}$; output-branch current $I_{D,out} \approx 83.6 \mu\text{A}$; and tail current $I_{tail} \approx 79.8 \mu\text{A}$. The output-branch-to-input current ratio gives the mirror gain

$$B = I_{D,out} / I_{D1} \approx 83.6 \mu A / 39.9 \mu A \approx 2.1 \quad (10)$$

Inverting the gain relation of Eq. (2) with the simulated $A_0 = 24.85 \text{ dB}$ ($= 17.5 \text{ V/V}$) yields the effective output resistance implied by the design,

$$R_{out} = A_0 / (B g_{m1}) = 17.5 / (2.1 \times 535 \mu S) \approx 15.6 \text{ k}\Omega \quad (11)$$

which corresponds to a per-device output resistance $r_o \approx 2R_{out} \approx 31 \text{ k}\Omega$ and an intrinsic device gain $g_{m1}r_o \approx 17$ — a physically reasonable value for a 2 V 180 nm-class device with the chosen 400 nm channel length. Substituting R_{out} and the unity-gain frequency into Eq. (5) back-solves the effective load capacitance:

$$C_L = B g_{m1} / (2\pi f_u) = (2.1 \times 535 \mu S) / (2\pi \times 14.6 \text{ MHz}) \approx 12.2 \text{ pF} \quad (12)$$

Independently, the measured slew rate fixes the load through Eq. (14): $C_L = B I_{tail} / \text{SR} = (2.1 \times 79.8 \mu A) / (15.7 \text{ V}/\mu s) \approx 10.7 \text{ pF}$. The two estimates- 12.2 pF from the small-signal bandwidth and 10.7 pF from the large-signal slew- agree to within $\approx 12 \%$, the residual being the expected slewing-plus-settling overhead. This close agreement across three independent measurements (gain, bandwidth, slew) confirms the model is self-consistent and validates the corrected slew relation $\text{SR} = B I_{tail} / C_L$ (using I_{tail} alone would predict $\approx 5 \text{ pF}$, contradicting the bandwidth result).

Quantity	Hand calculation	Simulated	Agreement
Mirror ratio B	$83.6 / 39.9 \approx 2.1$	m=4 vs m=2 (≈ 2)	consistent
DC gain A_0	$B g_{m1} R_{out} \approx 17.5 \text{ V/V}$	24.85 dB = 17.5 V/V	exact
Implied R_{out}	$A_0 / (B g_{m1}) \approx 15.6 \text{ k}\Omega$	Corresponds to ($g_{m1} r_o \approx 17$)	Physically reasonable
C_L from f_u	$B g_{m1} / 2\pi f_u \approx 12.2 \text{ pF}$	Derived from unity-gain bandwidth	Cross-check confirmed
C_L from SR	$B I_{tail} / \text{SR} \approx 10.7 \text{ pF}$	Derived from transient slew-rate result	Within approximately 12%

Table 4. Analytical verification- hand calculations from the operating point reproduce the simulated gain, bandwidth, and slew rate.

This closed-form reconciliation demonstrates that the simulated performance is grounded in the intended device operating point: the gain follows from the input transconductance and output resistance, the bandwidth from the same transconductance and the load, and the slew from the mirror-scaled tail current- all three pointing consistently to a ≈ 11 –12 pF effective load.

DC analysis confirms that every transistor sits in its intended region- saturation for the gain devices- and that the bias currents distribute as designed. It is the prerequisite that makes the AC and transient results trustworthy, since an incorrect device region invalidates the small-signal model. The operating-point annotation reports each device's I_D , V_{GS} , V_{DS} , and g_m , from which the gain of Eq. (2) is validated. Each gain device must satisfy the saturation condition.

For NMOS devices, the saturation (active-region) condition is

$$V_{DS} \geq V_{ov}, V_{ov} = V_{GS} - V_{th} \quad (13)$$

For PMOS devices, the equivalent condition is

$$V_{SD} \geq V_{ov}, V_{ov} = V_{SG} - |V_{th}| \quad (14)$$

where V_{ov} is the overdrive voltage. In Cadence Virtuoso, this condition should be confirmed using the operating-point region flag together with V_{DS} , V_{DSAT} , V_{GS} , I_D , and g_m .

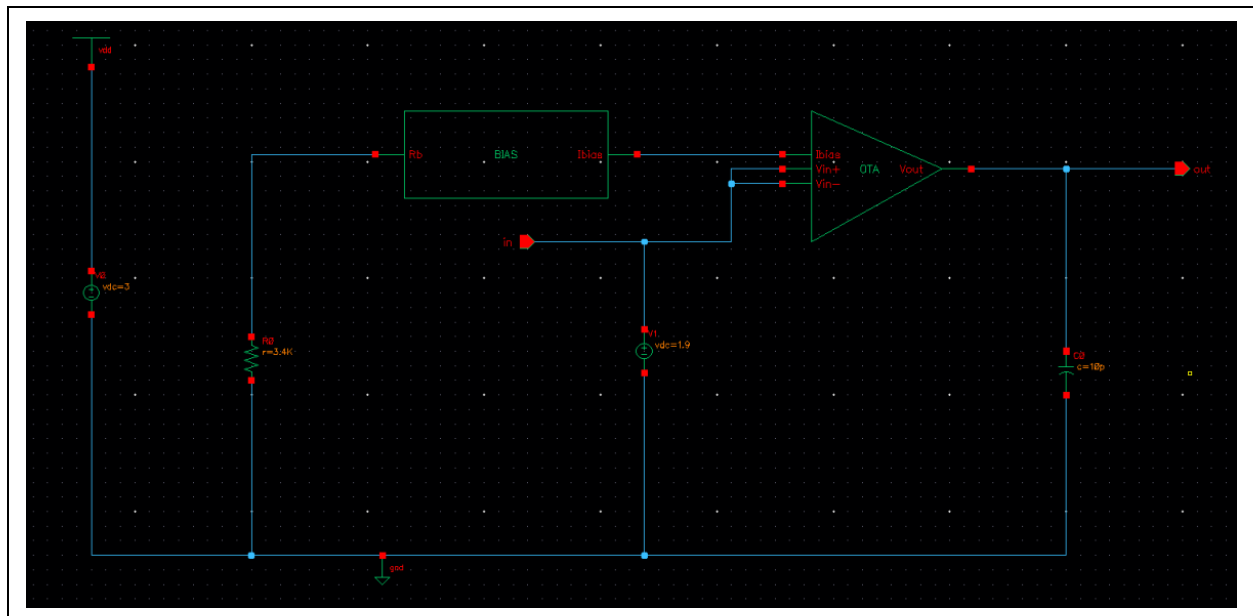


Figure 5. DC analysis test bench - used to bias the OTA and verify static node voltages and currents.

The diagram shows the DC test bench used to set the operating point and solve for every static node voltage and branch current. This step must precede AC and transient interpretation: the small-signal parameters g_m and r_o that determine gain and bandwidth are themselves functions of the DC bias, so a wrong operating point silently invalidates every subsequent plot even when those plots look plausible.

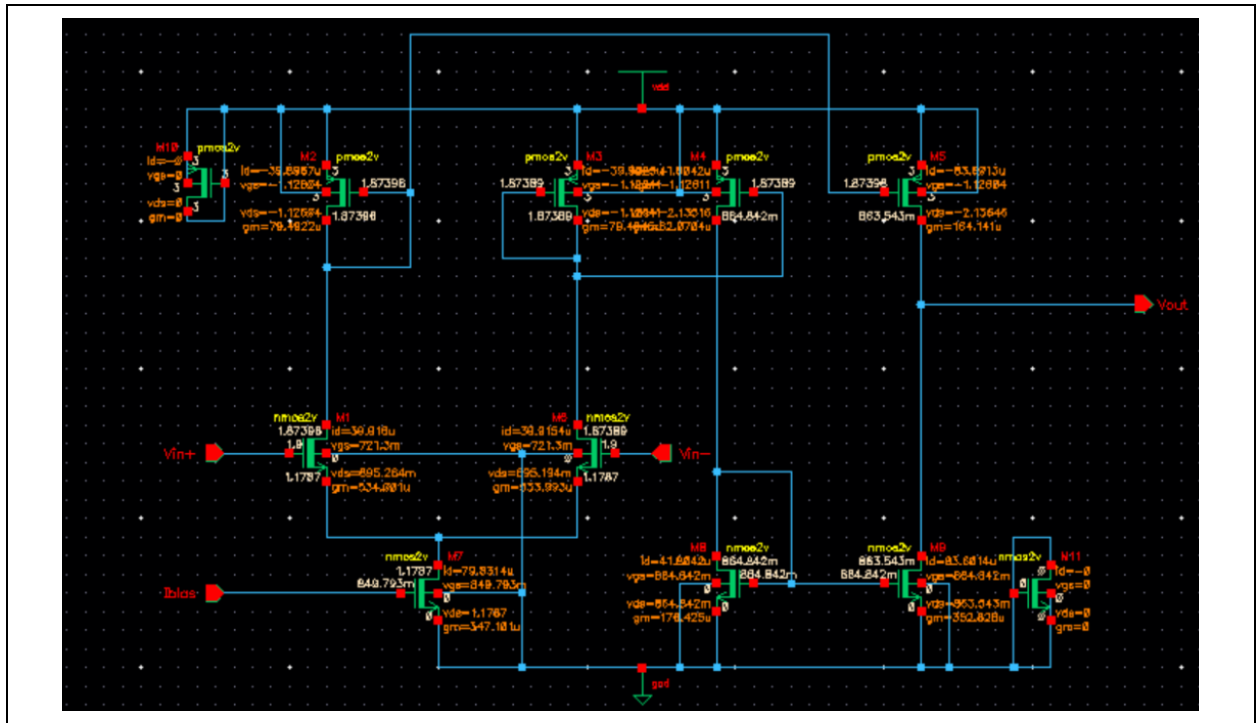


Figure 6. DC operating-point annotations — per-device branch currents and small-signal parameters on the schematic.

The diagram shows the operating-point annotations overlaid on the schematic - each device's branch current and small-signal parameters. The annotated currents confirm the mirror legs carry the intended ratios and that the input pair and output devices satisfy the saturation condition of Eq. (13) with adequate overdrive margin. In a formal sign-off these are cross-checked device by device against the Cadence OP report (region flag, V_{DS} vs. V_{DSAT} , and g_m/I_D), since a device biased only marginally in saturation degrades gain and adds distortion long before it fully leaves the region.

DC item	Verified Parameters	Design Significance
Input pair	ID, gm, VGS, VDS, region	Defines input transconductance, input linearity, and small-signal gain contribution
Current mirrors	Branch-current matching, saturation	Ensures accurate current steering, gain balance, and mirror-ratio consistency
Tail source	Current level	Sets OTA bias current, effective (g_m), slew-rate capability, and common-mode behaviour
Output branch	Output-node bias, saturation margin	Determines output swing, DC gain, distortion, and load-driving behaviour

Table 5. DC operating-point verification parameters and their design significance.

5. Large-Signal (Transient) Verification

Transient analysis exposes the large-signal limits that AC analysis cannot. When the input step is large enough to fully steer the tail current, the input pair acts as a hard current switch and the output slews at a rate set by the mirror-scaled tail current charging the load capacitance:

$$SR = dV_{out}/dt|_{max} = B I_{tail} / C_L \quad (15)$$

Slew rate and small-signal bandwidth share the same numerator: substituting $g_{m1} = 2I_{D1}/V_{ov1} = I_{tail}/V_{ov1}$ into Eq. (2) gives the identity $SR = 2\pi f_u V_{ov1}$, which states that, for a fixed unity-gain bandwidth, a larger input overdrive yields a proportionally faster slew. This couples the small- and large-signal design: choosing V_{ov1} trades input linearity and noise against slewing speed. The settling time to within an error band δ then adds the linear (exponential) settling governed by the closed-loop time constant:

$$t_s \approx t_{slew} + \tau \ln(1/\delta), \quad \tau = 1 / (2\pi f_u) \quad (16)$$

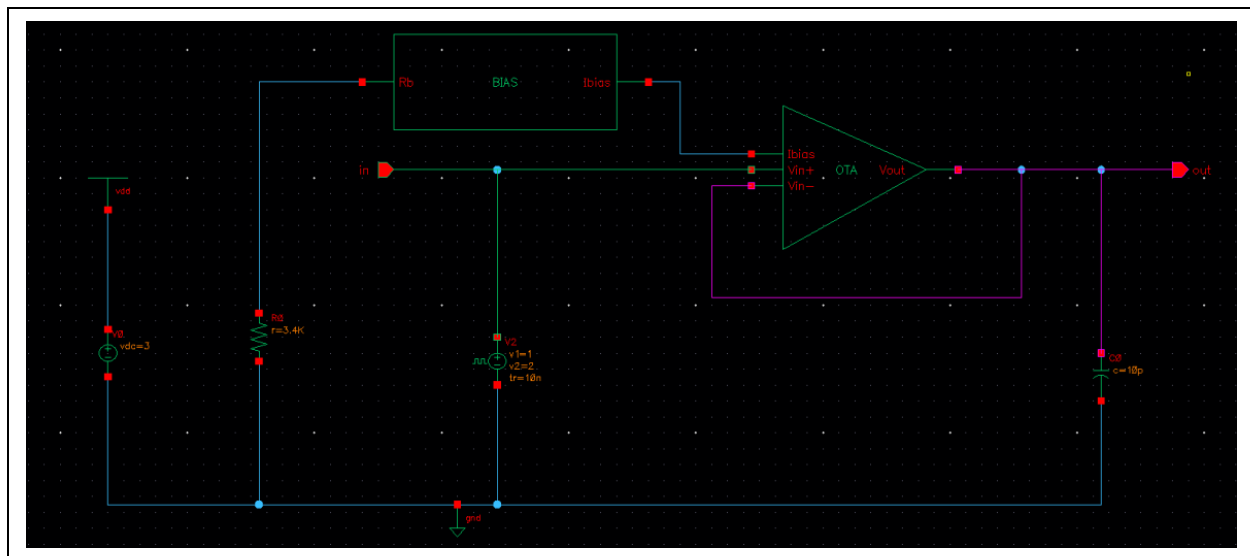


Figure 7. Transient test bench- OTA in unity-gain (buffer) configuration with a step input and load capacitor.

The diagram shows the OTA wired as a unity-gain buffer with a step input and load capacitor. This closed-loop configuration measures the large-signal response- slew rate and settling- under the same feedback condition the amplifier would see in use, complementing the open-loop AC test. The buffer is the most demanding stability case because the feedback factor is unity, so a design that settles cleanly here is safe at any lower closed-loop gain.



Figure 8. Slew rate $\approx 15.7 \text{ V}/\mu\text{s}$ (rising-edge dV/dt).

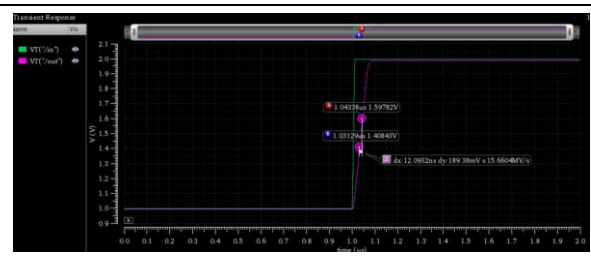


Figure 9. Settling response — no visible overshoot.

Parameter	Value	Notes
Rising slew rate	$\approx 15.7 \text{ V}/\mu\text{s}$	From dV/dt , Eq. (14)
Settling behaviour	Clean, no overshoot	Consistent with $\approx 73^\circ$ PM
Closed-loop stability	Stable buffer	Safe for feedback use

Table 6. Large-signal transient results.

The waveforms show the rising-edge slope and the step settling. The measured $15.7 \text{ V}/\mu\text{s}$ slew rate is set by the tail current charging the load (Eq. 15); because slew is current-limited rather than bandwidth-limited, it can only be raised by increasing I_{tail} or reducing C_L , which is why it is reported separately from the small-signal bandwidth. The settling waveform reaches its final value with no visible overshoot or ringing, the time-domain confirmation of the $\approx 73^\circ$ phase margin: a margin below $\approx 45^\circ$ would have shown overshoot here. The slewing and linear-settling phases together follow Eq. (16), and their clean combination confirms the amplifier is simultaneously fast and well-damped.

6. Common-Mode Input and Output Characterization

Gain, bandwidth, and slew rate are meaningful only within the input and output ranges where the transistor stack retains headroom. The input common-mode range (ICMR) is bounded below by the tail and input-pair headroom, and above by the load and input-pair saturation:

$$V_{ICM,min} = V_{DS,tail(sat)} + V_{GS,1} = V_{DS,tail(sat)} + V_{ov,1} + V_{th,1} \quad (17)$$

$$V_{ICM,max} = V_{DD} - |V_{DS,load(sat)}| - V_{ov,1} + V_{th,1} \quad (18)$$

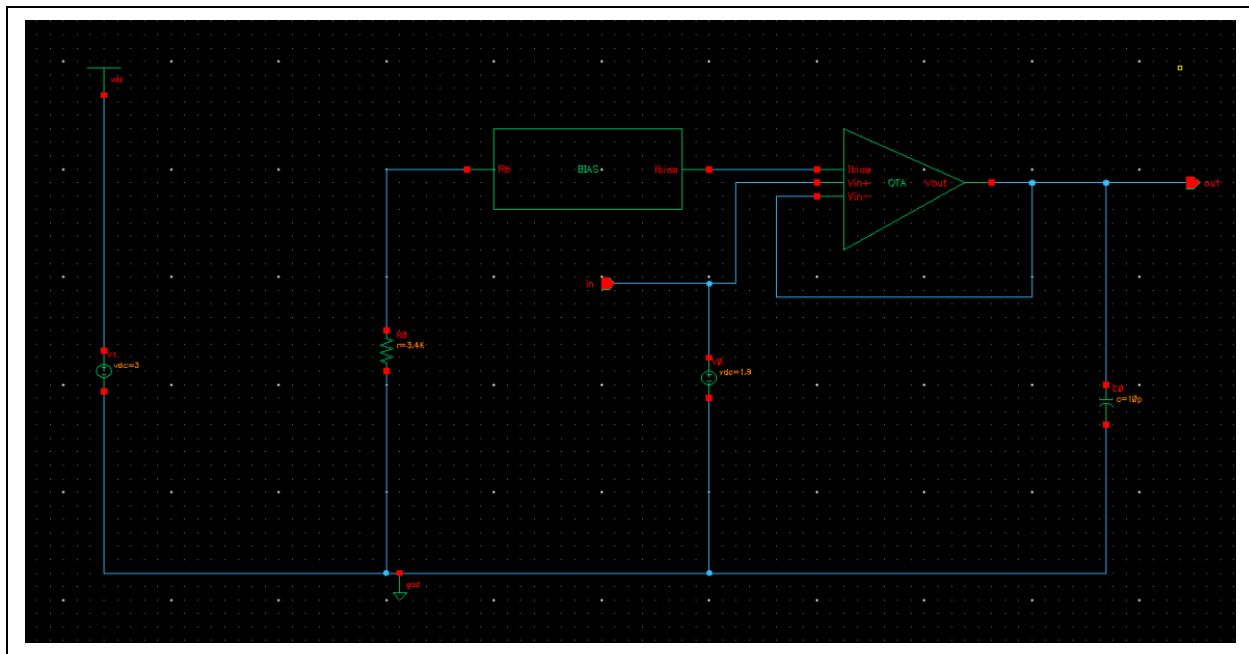


Figure 10. Common-mode input test bench — DC sweep of the common-mode input while monitoring OTA behaviour.

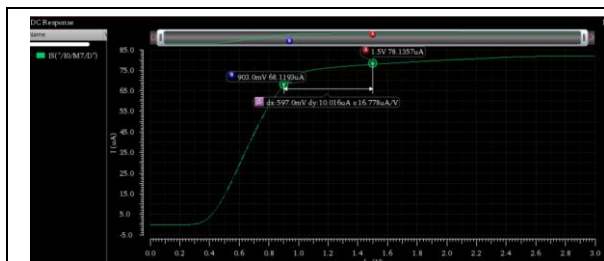


Figure 11. Lower ICMR boundary sweep.

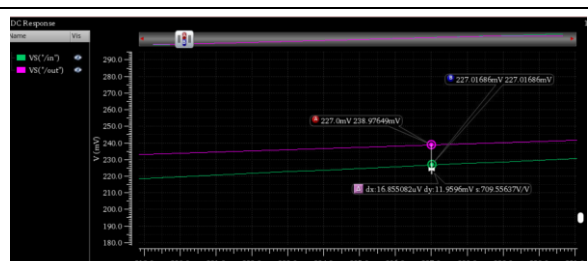


Figure 12. Upper ICMR boundary sweep.

The plots show the lower and upper common-mode input boundaries. The lower limit (Eq. 17) is reached when the tail source loses saturation- the input pair and tail can no longer both hold their overdrive- and below it g_m and gain collapse. The upper limit (Eq. 18) is reached when the PMOS loads or the input pair enter triode. The usable ICMR is the span between these, and it directly constrains how the amplifier may be biased in a system; an NMOS-input pair, as used here, naturally favours a common-mode level in the lower half of the supply, which is the expected behaviour seen in the sweeps.

The output swing is similarly bounded by the saturation margins of the output branch:

$$V_{out,min} = V_{DS,n(sat)}, \quad V_{out,max} = V_{DD} - |V_{DS,p(sat)}| \quad (19)$$

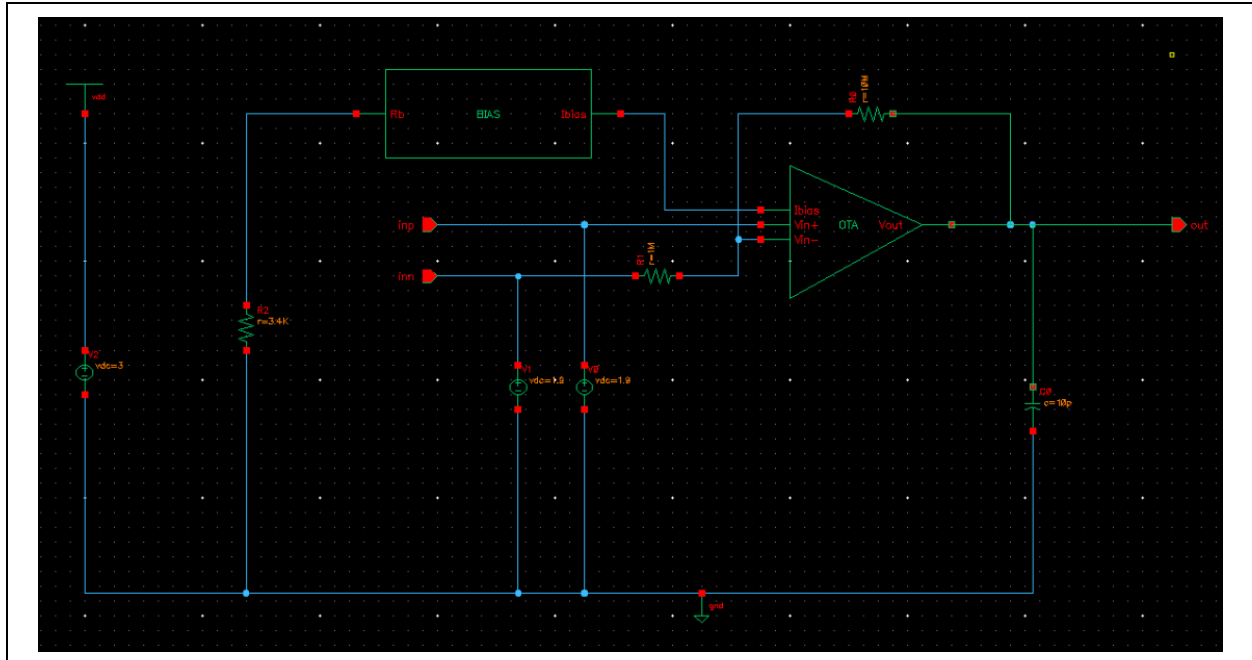


Figure 13. Common-mode output test bench — evaluates output-swing limits under swept conditions.

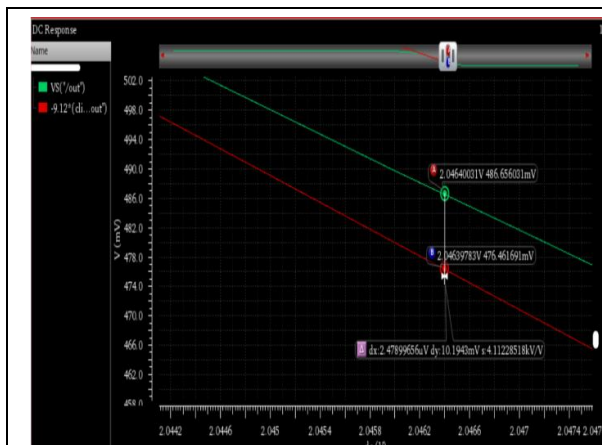


Figure 14. Lower output-swing boundary.

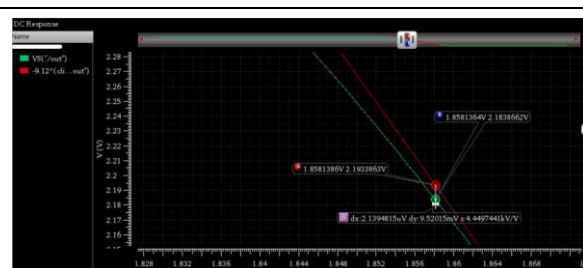


Figure 15. Upper output-swing boundary.

The plots show the lower and upper output-swing boundaries. The output remains linear only while both output-branch devices stay saturated, so the swing is bounded by their saturation voltages (Eq. 19): the NMOS side sets the lower limit and the PMOS side the upper. Characterizing this range alongside ICMR is what makes the specification honest - the quoted 24.85 dB gain applies only inside this window, and quoting gain without the accompanying swing would overstate the usable performance. The relatively wide swing of a single-stage non-cascoded output is, in fact, one of the advantages traded for its moderate gain.

7. Physical Layout, DRC and LVS Verification

Layout converts the verified schematic into a manufacturable geometry. In analog design it is not merely geometry: placement and routing directly affect offset, matching, parasitics, bandwidth, and noise. The devices were drawn with matching-aware techniques- common-centroid / interdigitated placement, dummy devices at array edges, and symmetric routing- so that the silicon reproduces the matching the schematic assumes (Eq. 4). Two sign-off checks gate any tape-out: the design-rule check (DRC), which enforces foundry geometric rules, and layout-versus-schematic (LVS), which confirms the extracted layout netlist is electrically identical to the schematic.

7.1 OTA layout and verification

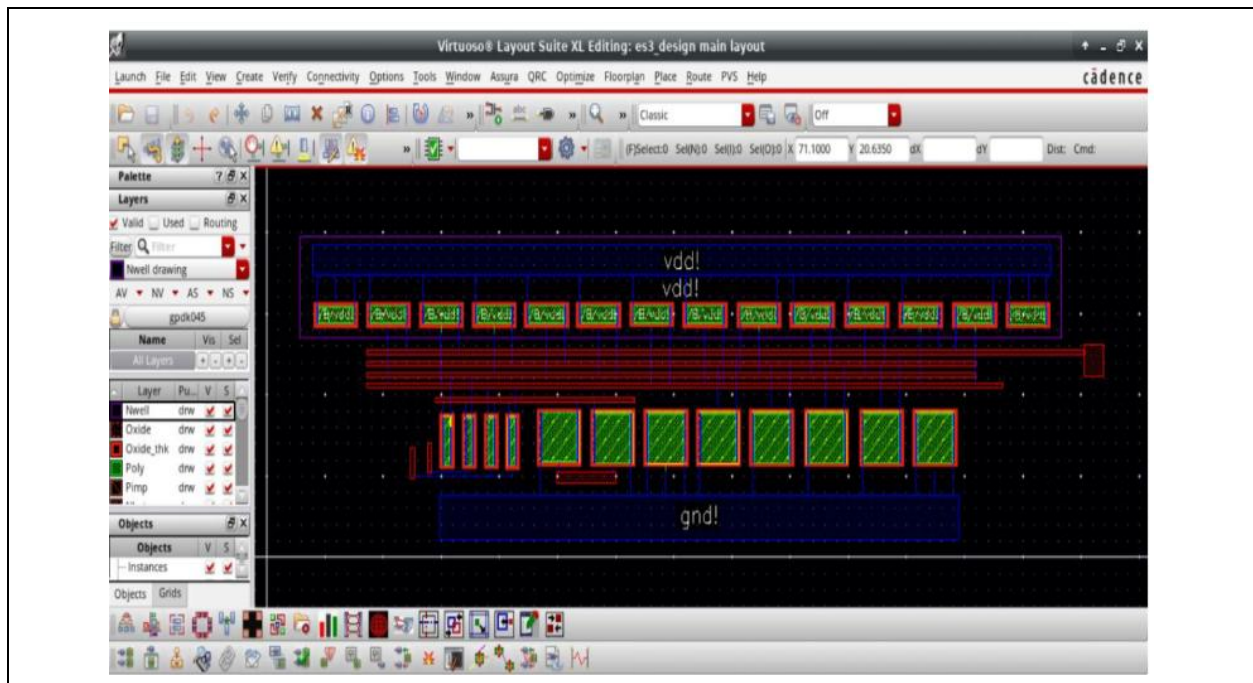


Figure 16. OTA core layout- matched device arrays, supply/ground rails, symmetric routing, and dummy devices.

The diagram shows the OTA core layout with its matched device arrays, supply and ground rails, and edge dummy devices. The symmetric, common-centroid arrangement cancels first-order process gradients (oxide thickness, doping, stress) across the matched pairs, directly reducing the systematic component of the input offset in Eq. (4); the random component is then set by the device area WL . The dummy devices at the array edges ensure every active transistor sees an identical neighbourhood, removing the etch and stress asymmetry that would otherwise appear at a boundary device. This is the step where the matching assumed in the schematic is actually earned in silicon.

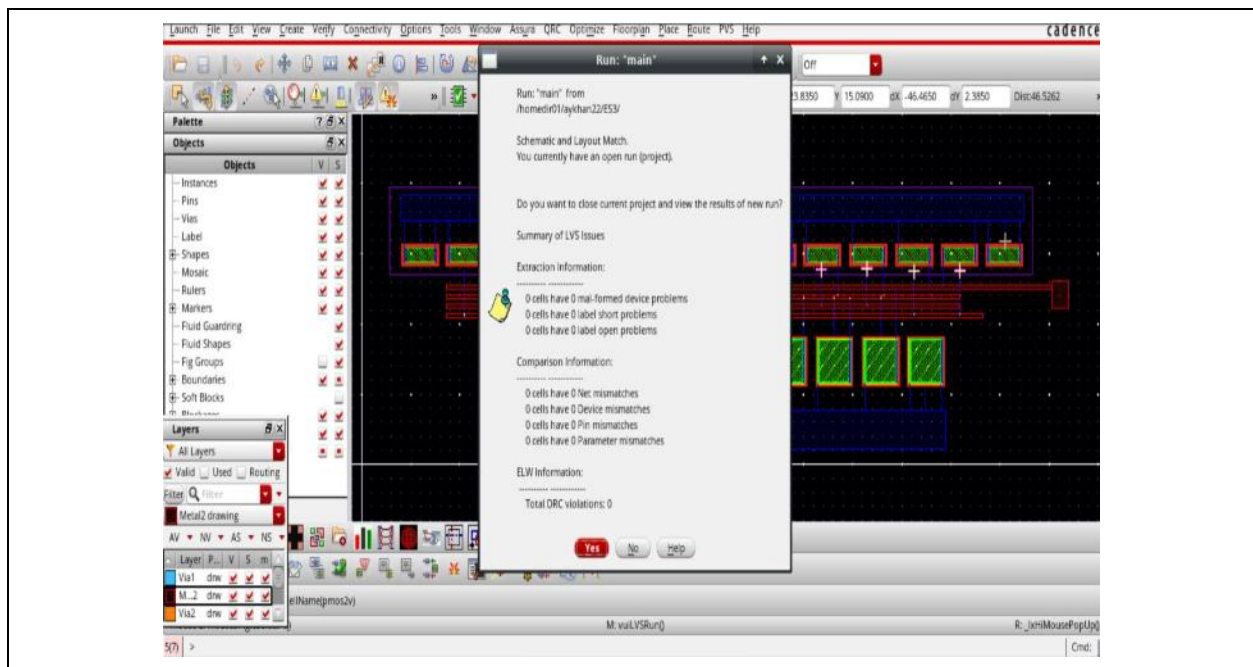


Figure 17. LVS for the OTA- extracted layout netlist matches the schematic (0 mismatches).

7.2 Bias-network layout and full integration

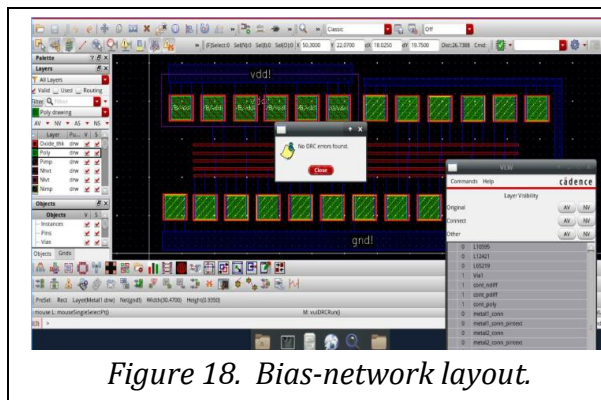


Figure 18. Bias-network layout.

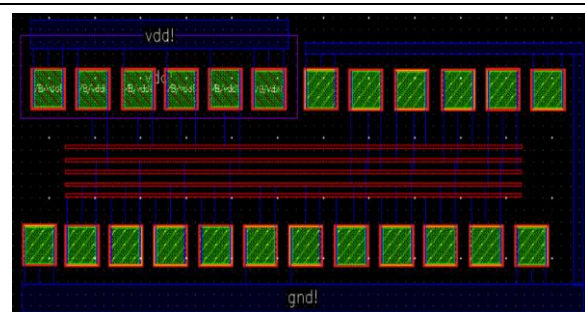


Figure 19. DRC of the bias circuit (0 violations).

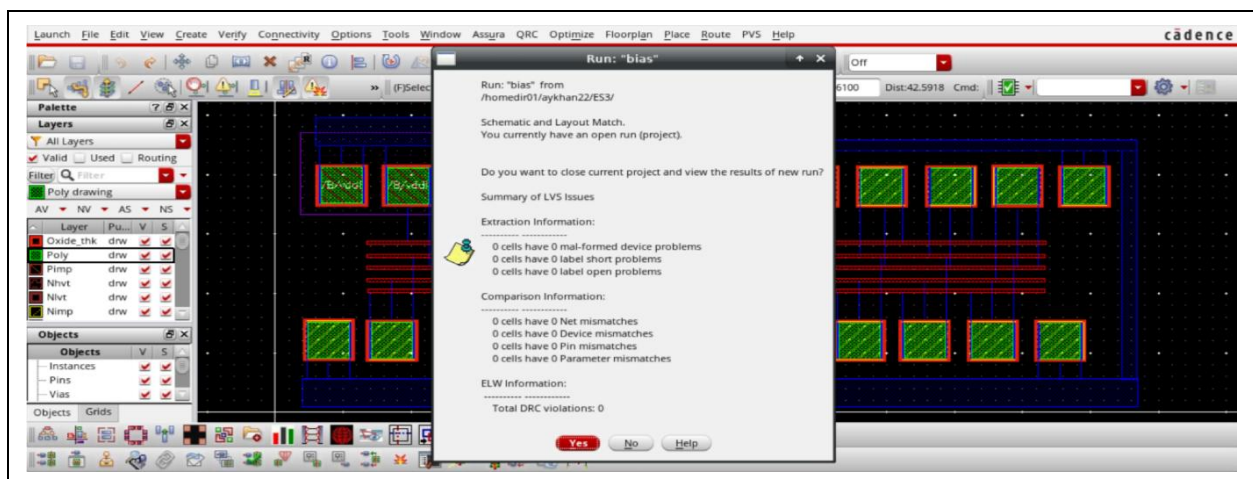


Figure 20. LVS for the bias circuit- schematic and layout match.

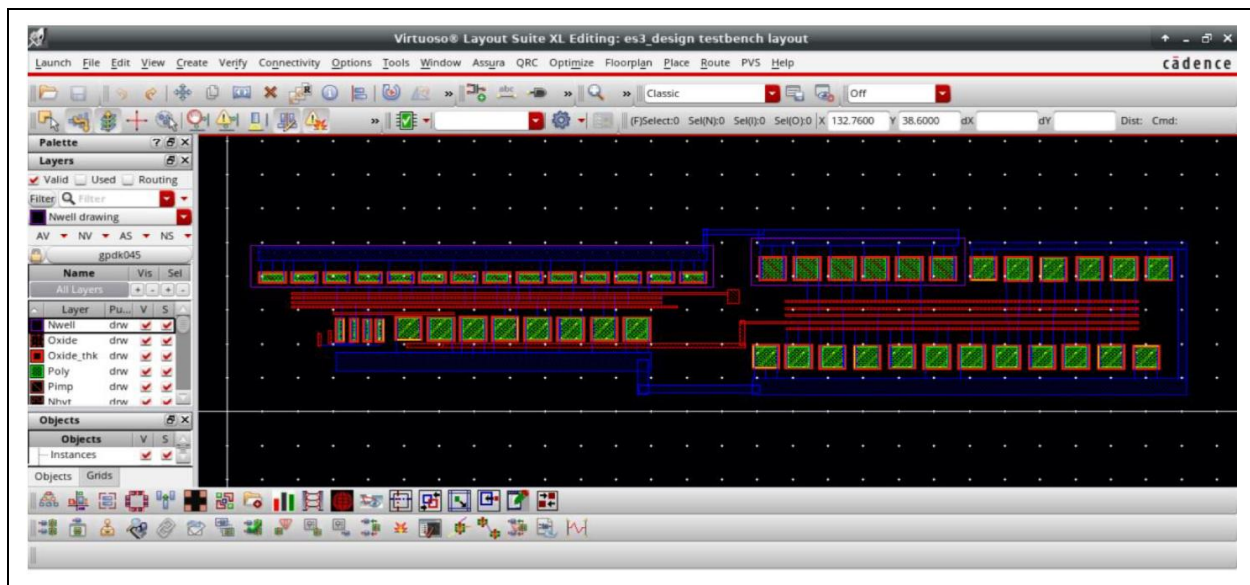


Figure 21. Full-circuit layout- OTA core and bias network integrated.

Check	Result	Meaning
DRC (OTA + bias)	0 violations	Geometry obeys all foundry rules
LVS (OTA + bias)	0 mismatches	Layout is electrically identical to schematic

Table 7. Physical-verification (sign-off) outcome.

The verification dialogs show zero DRC violations and a clean LVS match for both blocks. DRC confirms the geometry obeys every foundry rule (spacing, width, enclosure, antenna), while LVS confirms the extracted layout netlist is electrically identical to the schematic, device for device and net for net. A database that is clean on both is the minimum bar for tape-out; reaching it on the OTA and the bias network independently, before integration, is what separates a manufacturable design from a schematic-only exercise and is the discipline expected in any production analog flow.

8. Post-Layout Simulation and Parasitic-Aware Validation

Layout adds parasitic resistance and capacitance that the schematic cannot represent. Post-layout (extracted) simulation re-runs the key analyses on a netlist including these parasitics, answering whether the implemented circuit still meets specification. The principal effect is added capacitance at the output node, which lowers the dominant pole and the unity-gain bandwidth:

$$f_{u,PL} = B g_{m1} / [2\pi (C_L + C_{par})] \quad (20)$$

so a small downward shift in f_u is expected; the design retains margin if the shift is small. Figures 22–24 show the post-layout frequency response, bandwidth markers, and slew rate.

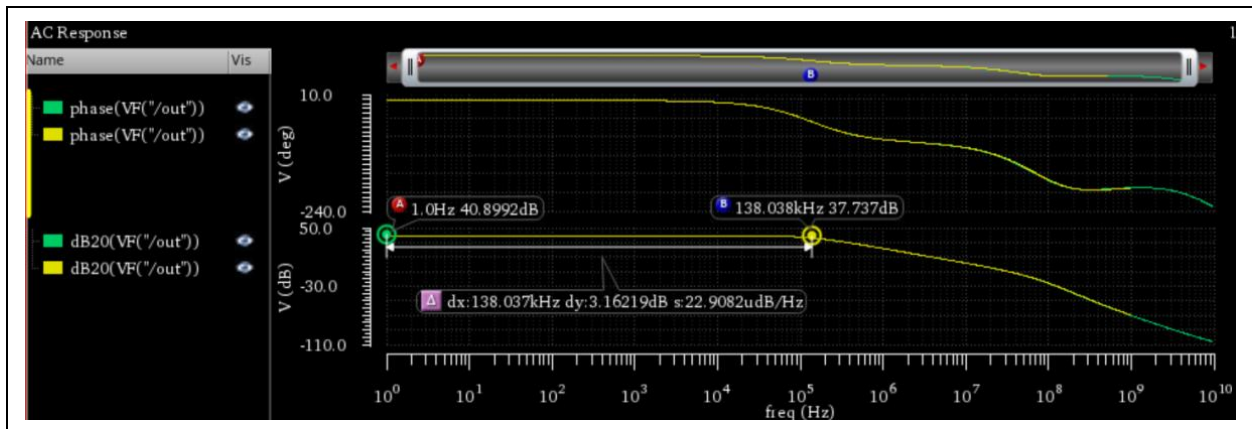


Figure 22. Post-layout frequency response — magnitude and phase after parasitic extraction.

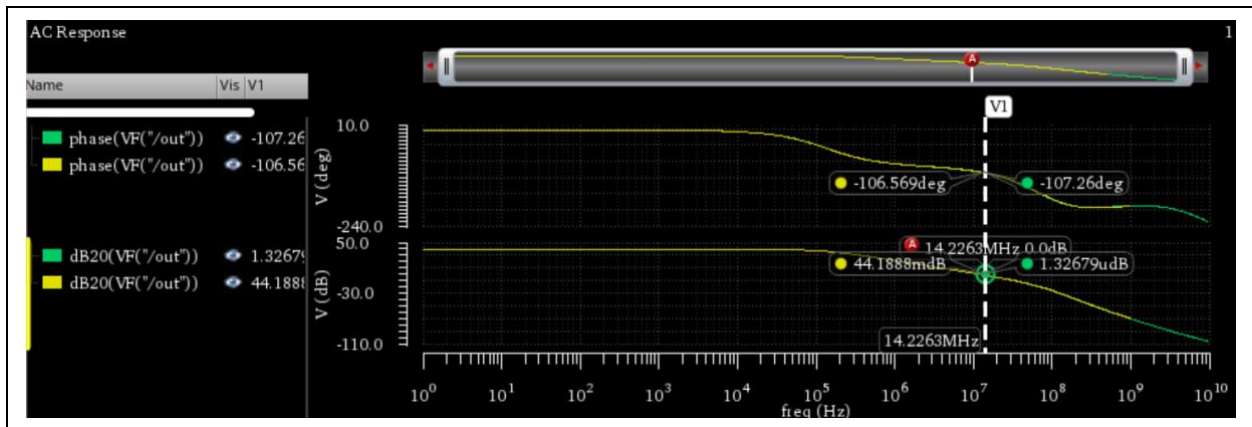


Figure 23. Post-layout bandwidth and 3 dB markers.

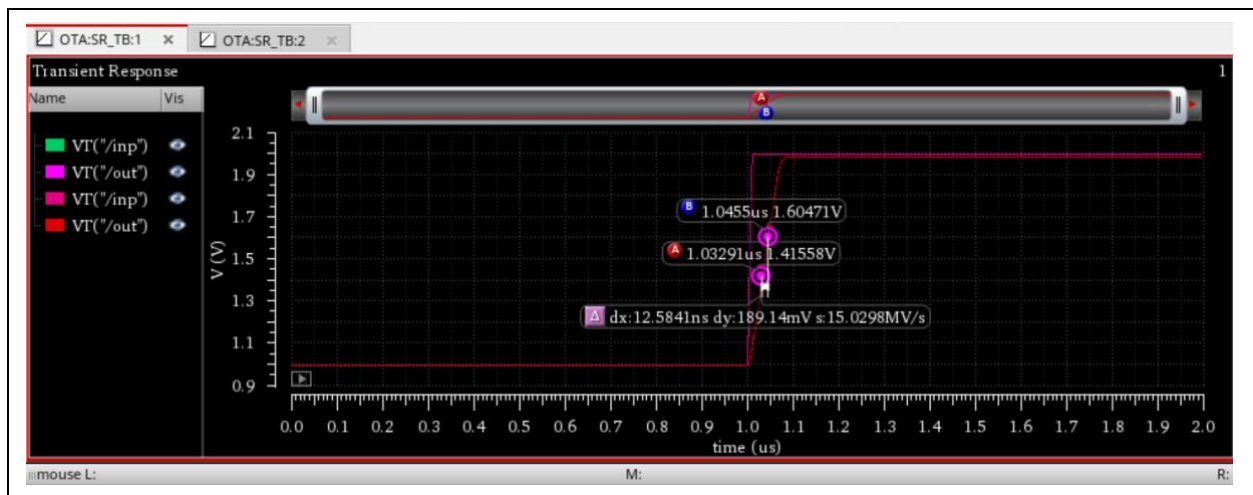


Figure 24. Post-layout slew-rate verification- extracted rising-edge transient.

The plots show the extracted-netlist response. The magnitude and phase retain the same dominant-pole shape as the schematic, and the bandwidth and slew rate sit close to their pre-layout values. The small downward shift in f_u is exactly what Eq. (20) predicts: the extracted interconnect and junction capacitance add to C_L at the output node, lowering the dominant pole. The fact that phase margin is preserved and the targets are still met demonstrates that the layout added little parasitic loading relative to the design budget- the practical payoff of the compact, single-high-impedance-node topology, whose one critical node is easy to keep low-capacitance. This parasitic-aware closure is the final evidence that the silicon implementation, not just the idealized schematic, meets specification.

9. Robustness Considerations: Process, Voltage, Temperature and Mismatch

Beyond the nominal operating point, a production-grade analog block must hold its specification across process corners, supply and temperature variation (PVT), and random device mismatch. This section sets out the analytical sensitivities and the verification methodology that govern this robustness. The relations below are derived from the design equations of Sections 2–3; the explicit Monte-Carlo and corner simulations they motivate were beyond the scope of the present nominal-design study and are identified as the natural next verification step.

9.1 Sensitivity to process and bias variation

The two headline specifications depend on the bias-derived small-signal parameters, so their first-order sensitivities follow directly from Eqs. (1)–(3). Since $g_m = \sqrt{2\mu_n C_{ox}(W/L)I_D}$, a fractional change in either the process factor $\mu_n C_{ox}$ or the bias current I_D propagates to the transconductance as

$$dg_m / g_m = \frac{1}{2} (d(\mu_n C_{ox}) / \mu_n C_{ox}) + \frac{1}{2} (dI_D / I_D) \quad (21)$$

so a $\pm 20\%$ swing in either quantity moves g_m by only $\approx \pm 10\%$ (the square-root compresses the variation). The unity-gain bandwidth $f_u = Bg_{m1}/2\pi C_L$ tracks g_{m1} linearly and therefore shares this $\pm 10\%$ sensitivity, while the DC gain $A_0 = Bg_{m1}R_{out}$ is more variable because both g_{m1} and $R_{out} \propto 1/(\lambda I_D)$ move with process. Combining the two,

$$dA_0 / A_0 = dg_{m1}/g_{m1} + dR_{out}/R_{out} \approx \frac{1}{2} dI_D/I_D - dI_D/I_D = -\frac{1}{2} dI_D/I_D \quad (22)$$

which shows the useful result that, to first order, gain *falls* with increasing bias current while bandwidth *rises*—the classic analog gain–bandwidth trade. The two compensate partially, which is why the gain–bandwidth product (Eq. 5) is the more stable figure of merit across process.

9.2 Corner behaviour

Table 8 summarizes the expected direction of each specification at the slow (SS), typical (TT), and fast (FF) corners, derived from the sensitivities above. The slow corner—low mobility, high threshold—is worst-case for gain and bandwidth; the fast corner is worst-case for power and may compress the input common-mode range as thresholds drop. A correct sign-off would re-run the AC, transient, and ICMR analyses of Sections 3–6 at each corner and confirm the phase margin stays above 60° throughout.

Specification	Slow (SS)	Typical (TT)	Fast (FF)
Transconductance g_m	↓ (low μ_n , high V_{th})	nominal	↑ (high μ_n)
DC gain A_0	↑ (higher r_o)	≈ 24.85 dB	↓ (lower r_o)
Unity-gain BW f_u	↓ (low g_m)	≈ 14.6 MHz	↑ (high g_m)
Phase margin	↑ (lower f_u)	$\approx 73^\circ$	↓ (higher f_u) - check
Slew rate	↓	≈ 15.7 V/ μ s	↑
ICMR	narrows (high V_{th})	nominal	check upper limit

Table 8. Predicted direction of each specification across process corners (analytical projection).

9.3 Mismatch and input-referred offset

Random device mismatch sets the input-referred DC offset, which for this topology is dominated by the threshold mismatch of the input pair and the current-factor mismatch of the load mirrors. Referring both contributions to the input gives

$$\sigma_{Vos}^2 = \sigma_{\Delta V_{th,1}}^2 + (g_{m,p} / g_{m1})^2 \sigma_{\Delta V_{th,p}}^2 \quad (23)$$

where the first term is the input-pair contribution and the second the mirror-load contribution weighted by the transconductance ratio. Two design choices already in the schematic directly minimize this offset: biasing the input pair at high g_{m1} relative to the load $g_{m,p}$ suppresses the second term, and (via Pelgrom's law, Eq. 4) the chosen large input-device area WL reduces $\sigma_{\Delta V_{th}}$. Substituting Eq. (4) makes the area dependence explicit:

$$\sigma_{Vos} = \sqrt{[A_{V_{th,n}}^2 / (W_1 L_1) + (g_{m,p} / g_{m1})^2 A_{V_{th,p}}^2 / (W_p L_p)]} \quad (24)$$

This is the relation a Monte-Carlo mismatch simulation would sample statistically: each run perturbs every device's V_{th} and current factor by independent Gaussian deviates and reports the resulting offset distribution, from which the 3σ offset is taken as the worst-case specification. The common-centroid layout of Figure 16 additionally cancels the systematic (gradient-induced) component, leaving only this random term. A complete characterization would report the simulated mean and 3σ offset against the budget implied by Eq. (24); the present work establishes the analytical target and the layout measures that make it achievable.

10. Consolidated Results and Discussion

Table 9 consolidates the schematic-level and post-layout results against the design goals of Section 1.1. Every target is met: the amplifier is unconditionally stable ($PM \approx 73^\circ \gg 60^\circ$), reaches a unity-gain bandwidth in the tens of MHz, and is DRC/LVS-clean with parasitic-aware behaviour confirmed.

Metric	Schematic	Post-layout
DC gain, A_0	≈ 24.85 dB	Preserved; extracted AC response keeps the same low-frequency gain region within design margin
Unity-gain bandwidth	≈ 14.6 MHz	Close to schematic; small downward shift expected from extracted C_{par} , Eq. (19)
Phase margin	$\approx 73^\circ$	Stable dominant-pole behavior; no peaking or oscillation visible in extracted response
Slew rate	≈ 15.7 V/ μ s	Close to schematic; extracted transient preserves large-signal speed
DRC / LVS	0 / 0	Clean for OTA core and bias network; 0 violations / 0 mismatches

Table 9. Schematic vs. post-layout results.

10.1 Skills and Competencies Demonstrated

This project exercised the complete full-custom analog IC competency set. It demonstrated the Cadence Virtuoso flow end to end- schematic capture, symbol and test-bench creation, simulation, layout, DRC, LVS, and post-layout verification. At the transistor level it required symmetrical current-mirror OTA design, biasing, operating-point design, and gm/Id reasoning, together with differential-pair, active-load, and output-headroom analysis. The verification methodology spanned small-signal AC (gain, bandwidth, phase margin), DC operating point, large-signal transient (slew and settling), common-mode range, and parasitic-extracted post-layout analysis. The physical work applied layout-aware design- dummy devices, symmetric placement, matched arrays, and routing discipline- and the sign-off discipline of DRC/LVS-clean databases and parasitic-aware validation. Throughout, results were interpreted against closed-form relations and reported with an honest distinction between values read from plots and exact simulator data.

11. Measurement Methodology and Bench Correspondence

Although this project is a simulation- and layout-stage design and the OTA was not physically fabricated, each simulated analysis has a direct hardware-measurement counterpart. This section maps the simulated results in this report to the bench instruments and measurement techniques that would be used to characterize a fabricated prototype. It demonstrates that the design analysis is framed in measurable engineering terms and that the path from Cadence simulation to silicon validation is understood.

11.1 Simulation-to-instrument correspondence

Table 10 maps each simulated analysis to the corresponding bench instrument and validation method. The correspondence is based on the same physical quantities: AC simulation predicts gain, bandwidth, and phase behaviour; transient simulation predicts slew rate and settling response; DC operating-point analysis predicts bias currents and node voltages; and noise/linearity analysis can be validated using appropriate laboratory instrumentation.

Simulated analysis (this report)	Bench instrument	Measurement technique
AC magnitude / phase (Fig. 4)	Vector network analyzer (VNA)	S-parameters after SOLT/TRL calibration; gain, BW, phase margin from S21
DC operating point (Fig. 6)	SMU / DC source-meter/precision multimeter	I-V sweeps; bias current, node voltage, VGS, VDS verification
Slew / settling (Figs. 8–9)	Digital storage oscilloscope	Step response; dV/dt slew, settling-time cursor, overshoot check
Common-mode range (Figs. 11–15)	SMU + oscilloscope	DC sweep of input/output operating range and linear-range boundaries
Large-signal linearity	Signal generator + spectrum analyzer	CW or two-tone stimulus to observe harmonics, IM3 products, and IP3 estimation
Noise (analytical, Eq. 8)	Spectrum analyzer / NF meter	Output-noise spectral-density measurement and input-referred noise calculation

Table 10. Correspondence between the simulated analyses and the bench instruments that measure the same quantities.

11.2 Calibration and de-embedding

Accurate measurement requires removing test-setup errors before characterizing the device. For high-frequency, on-wafer, or fixture-based measurements, SOLT or TRL calibration can move the reference plane to the device terminals. Pad, probe, fixture, and interconnect parasitics should be removed using open/short or thru de-embedding structures.

For a 50 Ω RF measurement setup, the reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (25)$$

and the voltage gain is

$$A_v(\text{dB}) = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right| \quad (26)$$

where Z_0 is the 50Ω system reference impedance. Since an OTA is a high-impedance analog block, direct 50Ω loading may disturb the measurement. Therefore, a fabricated OTA should be measured using a suitable buffer, high-impedance probe, or dedicated test structure.

The low-frequency gain and bandwidth from the AC simulation in Figure 4 would be validated using a small-signal frequency-response measurement. Depending on the test structure, this could be measured using a frequency-response analyzer, oscilloscope-based gain/phase setup, or a VNA with proper buffering and de-embedding. The measured magnitude gives gain and bandwidth, while the measured phase or loop-gain response is used to estimate phase margin.

11.3 Time- and frequency-domain validation

In the time domain, an oscilloscope would measure the slew rate and settling response predicted by transient simulation. The slew rate is extracted as :

$$SR = \max \left(\frac{dV_{out}}{dt} \right) \quad (27)$$

In the frequency domain, gain, bandwidth, and phase can be measured using a frequency-response analyzer, network analyzer with proper buffering, or oscilloscope-based gain/phase setup. Linearity can be evaluated using a signal generator and spectrum analyzer to observe harmonic distortion and two-tone intermodulation products. These measurements connect the Cadence simulation results to silicon-level validation.

12. Conclusion

A single-stage symmetrical current-mirror CMOS OTA was designed and fully implemented in Cadence Virtuoso, in a 180 nm-class 2 V CMOS process, through the complete analog IC flow. The schematic-level design achieves ≈ 24.85 dB DC gain, ≈ 14.6 MHz unity-gain bandwidth, $\approx 73^\circ$ phase margin, and ≈ 15.7 V/ μ s slew rate, each interpreted against closed-form analog relations. A matching-aware layout employing dummy devices and symmetric routing was verified DRC- and LVS-clean for both the OTA core and the bias network, and post-layout simulation with extracted parasitics confirmed the schematic-level performance. The project demonstrates ownership of an analog block from transistor-level

concept to a manufacturable, verification-clean layout whose extracted behaviour matches intent- the core competency required for analog / RF integrated-circuit, mixed-signal, and PHY-layer hardware design.

13. References

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